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SIR-SRGAN-ResNeXt: A New Super-Resolution GAN with Self-Interpolation Ranker and ResNeXt Generator

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Abstract

Super-resolution can be a powerful tool in enhancing image quality and bringing clarity to image details. It is essential for various fields such as medical and surveillance imaging as it improves image resolution to reveal fine details. Super-Resolution Generative Adversarial Networks (SRGAN) have shown promising capabilities toward perfecting super-resolution. However, the SRGAN with Self-Interpolation Ranker (SIR-SRGAN), using the difference between the reconstructed and the original image, often produces images with blurred areas, compromising image quality. This paper introduces SIR-SRGAN-ResNeXt, an improved version of SIR-SRGAN capable of generating clearer images with higher-quality metrics. The proposed model retains the Self-Interpolation classifier of SIR-SRGAN, incorporates a U-net-based discriminator, and adds attention layers for more effective feature analysis. Moreover, the generator is shifted to a more complex ResNeXt-based model, resulting in improved performance when evaluated against state-of-the-art SRGAN models in terms of high resolution and optimal output file size.

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1. Introduction

Single-image super-resolution (SISR) is a computer vision task that reconstructs a high-resolution image from a low-resolution image. It is widely used in various applications such as medical imaging for earlier diagnosis and surveillance imaging. Designing an SISR model involves addressing two major challenges: firstly, the model must generate an image with good visual quality that is as close as possible to the original image; secondly, generated images should have minimal blurriness. Although GAN-based models, namely SRGANs [8], produce good-quality images, they still generate blurred areas. Rank-SRGAN [24], thanks to the use of a Self-Interpolation Ranker, and BSRGAN [22] have mitigated this issue by producing clearer images, but resulting images are a little further from

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real images (e.g., excessively smooth) even though they are clear. Another challenge with the SRGAN model involves accurately representing the contours of objects (high frequencies) as they are very blurred. With two Self-Interpolation Rankers, the SIR-SRGAN [4] model is an improvement on SRGAN and Rank-SRGAN so that generated images come the closest possible to real ones. In addition, the SIR-SRGAN model incorporates wavelet loss enhancing its ability to generate more precise contours, especially in high-frequency areas.

This paper introduces an extended iteration of the SIR-SRGAN model. While preserving the Self-Interpolation Ranker from SIR-SRGAN, various improvements have been implemented. Enhancements include replacing the discriminator based on the PatchGAN architecture with a U-Net-based discriminator. Attention layers were also added to this discriminator to compel the model to analyze extensive features more effectively. Furthermore, initially based on ResNet, the generator was replaced with a ResNeXt-based generator. This shift to ResNeXt allows for increased model complexity to capture more features without compromising the training stability. Experiments on medical and general-purpose images show that the use of a Standard Ranker and a Feature Ranker in conjunction with a ResNeXt-based generator and a U-Net discriminator marks a significant advance in capturing complex details and improving the quality of super-resolved images with optimal storage efficiency and relatively reasonable computation time.

The remainder of this paper is organized as follows: Section 2 gives a more detailed presentation of the related works. In Section 3, the proposed SIR-SRGAN-ResNeXt is thoroughly presented. Section 4 outlines the conducted experiments and provides corresponding discussion. Section 5 concludes the paper.

2. Related Works

This section describes the most popular and recent single-image super-resolution baseline GAN models. These baseline models are beneficial for ascertaining the reliability of the proposed SIR-SRGAN-ResNeXt model. The Super-Resolution Generative Adversarial Network (SRGAN) [8] is a network developed by Ledig et al. that uses an SRResNet architecture with residual connections to learn low-resolution features. It also utilizes a pre-trained VGG model to calculate content loss, resulting in more realistic and detailed visual results. RankSRGAN [24] is a model introduced by Zhang et al. that uses a CNN architecture called "a Ranker" to rank images based on their perceptual quality. The Ranker is trained to guide the generator in the direction of metrics using "rank-content loss", resulting in less blurred and more detailed images compared to SRGAN. It acts as a second discriminator, giving a score to each generated image and forcing the generator to orientate its training to obtain better scores. Proposed by Zhang et al., BSRGAN [22] is an image super-resolution model that includes a more sophisticated degradation model, namely randomly mixed blur, which yields better results than SRGAN on images with many visual degradations, particularly on human facial images. The model includes randomly mixed blur, subsampling operations, and noise degradations, with blur approximated by two convolutions with isotropic and anisotropic Gaussian kernels. Subsampling is performed through the nearest neighbor, bilinear interpolation, and cubic convolution. Noise is synthesized by adding Gaussian noise with varying levels and applying JPEG compression with different quality factors. BSRGAN uses a U-Net discriminator to improve the quality of the generated images. Schonfeld et al. [13] have demonstrated that U-Net discriminators, compared to classical discriminators, extract simultaneous global and local characteristics, providing coherent spatial feedback to the generator. In contrast, a classical discriminator extracts either local or global characteristics. The U-Net discriminator gives a more pronounced return to the generator than a regular discriminator. The Super-Resolution Generative Adversarial Network with Self-Interpolation Ranker (SIR-SRGAN) [4] is an image super-resolution method that focuses on the intrinsic differences between the reconstructed image and the original image. It has two Rankers, trains them simultaneously with the generator, and uses image interpolation to enrich the training data. SIR-SRGAN also adds a new component to the loss function, the Patch Distance Loss (PDL). It measures the distance between high-frequency patches in the image using cosine similarity to calculate the distance between patches. The Self-Attention Generative Adversarial Network (SAGAN) was introduced by Zhang et al. [21]. It is a Generative Adversarial Network that uses a self-attention mechanism [18] to improve the consistency and quality of generated images. It resolves spatial coherence and structure issues in conventional GAN models and has shown improved performance in generating realistic human facial images.

Juhong et al. propose SRGAN-ResNeXt [6], a variant of SRGAN with a generator that employs the ResNeXt architecture. ResNeXt introduces cardinality to improve performance without increasing the complexity of the model. Unlike ResNet, ResNeXt uses parallel residual blocks to capture richer features and improve model capacity without

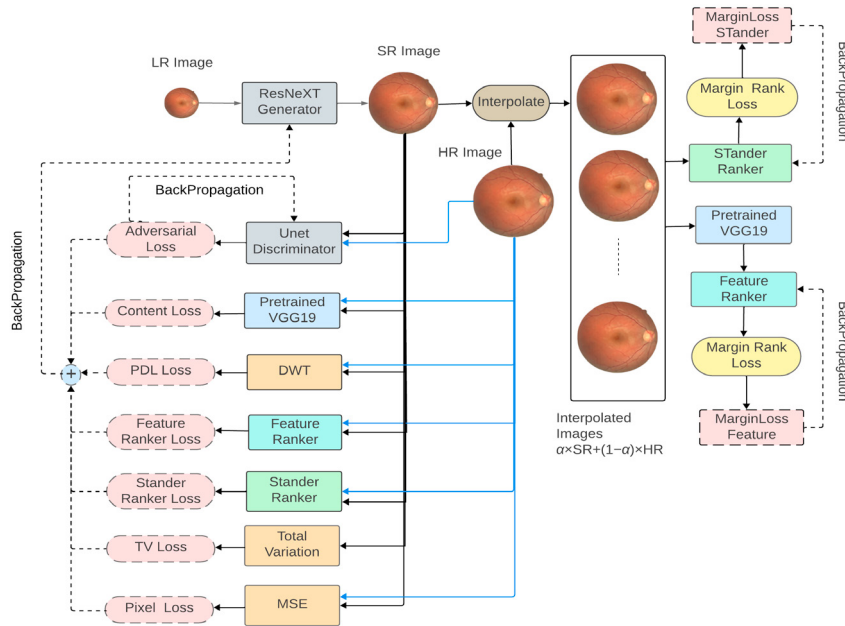


Fig. 1. Architecture of the proposed SIR-SRGAN-ResNeXt model.

overfitting. SRGAN-ResNeXt offers a significant improvement in image quality over SRGAN with ResNet, as shown in the study on breast cancer histopathological images.

3. The SIR-SRGAN-ResNeXt Model

The general architecture of the proposed SIR-SRGAN-ResNeXt is inspired by that of SIR-SRGAN. It is based on the use of two rankers, the Standard Ranker and the Feature Ranker, in conjunction with a generator and a discriminator. During training, image interpolation is integrated using images produced by the generator with different interpolation factors between reconstructed and high-resolution images. The Standard Ranker is designed to sort these interpolated images by learning the intrinsic differences between reconstructed and high-resolution images. At the same time, the Feature Ranker uses high-level features extracted by the VGG19 model [14] to evaluate these images. These rankers are important because they guide the generation process to improve the perceived quality of super-resolution images. The discriminator then evaluates the probability of the generated images, while the generator continuously tries to generate images that fool the discriminator.

This architecture also incorporates Patch Distance Loss, a metric derived from SIR-SRGAN, to properly represent high frequencies. The updates introduced in this architecture bring significant improvements to the super-resolution process. The standard generator has been replaced with a ResNeXt-based generator, providing increased capacity to learn more complex features and generate higher-quality super-resolution images. Similarly, the conventional discriminator has been replaced with a U-Net architecture, which promotes a more robust evaluation of the generated images and contributes to better discrimination between reconstructed and original images. Adding attention layers at the large layers of the discriminator allows the discriminator to optimize large feature extractions, to improve the model's ability to focus on key regions during the training process. The loss function has a component called Total Variation (TV loss) [12], which helps reduce noise in the image, particularly around the objects' edges in the image. Figure 1 presents the overall architecture of the proposed SIR-SRGAN-ResNeXt model.

3.1. Generator and Discriminator of the SIR-SRGAN-ResNeXt Model

The generator of the proposed model uses the ResNeXt architecture to learn deep and complex representations. Parallel convolution blocks allow greater model expressiveness, enabling it to learn more features without impacting

the training stability. Given its significant advantages, we incorporate the ResNeXt architecture into the proposed model's generator. This architecture is illustrated in Figure 2.

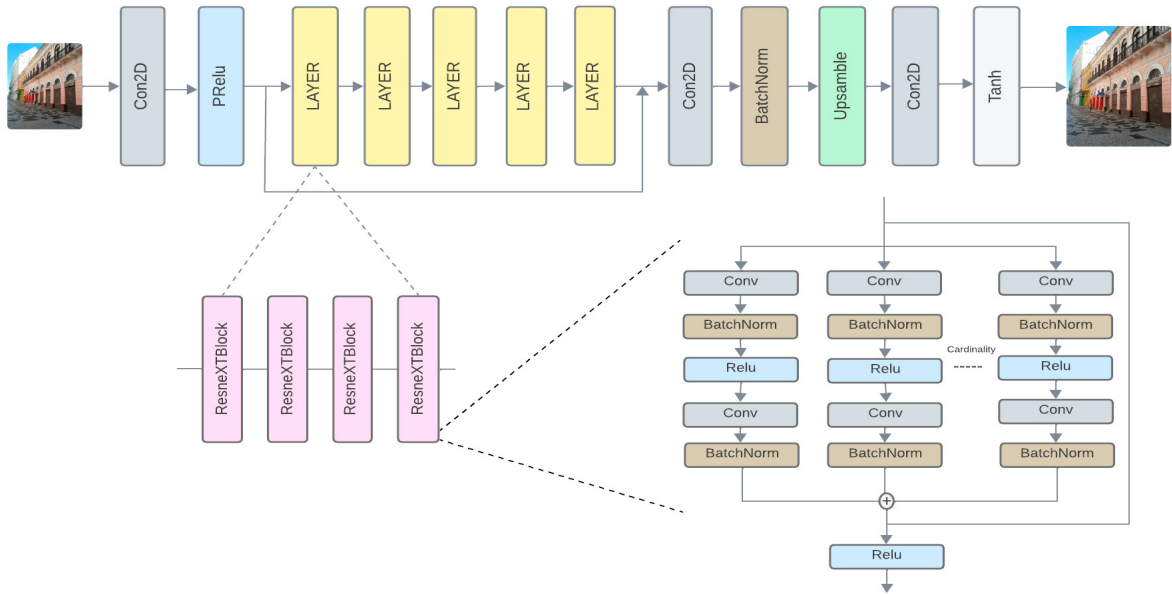


Fig. 2. Generator of the SIR-SRGAN-ResNeXt model.

On the other hand, the discriminator is built according to the U-Net architecture, illustrated in Figure 3, where spectral normalization [10] was applied to each of its layers. The model takes an image as input and uses multiple layers of bottom-up and top-down resolution with residual connections and attention mechanisms to extract features at different scales and restore spatial resolution. The model uses the Leaky ReLU [9] activation function and spectral normalization to stabilize the learning. Ultimately, the discriminator produces a single output value indicating the probability an input image is real. The model is designed to improve the stability of the learning and at the same time, the model's potential to discern fine details in the generated images.

3.2. Loss Function

The overall loss function of the proposed model is given in Equation 1:

$$loss_{SR} = \alpha \cdot l_{TVLoss} + \beta \cdot l_{VGG} + \gamma \cdot l_{Adv} + \lambda \cdot l_{Rank} + \theta \cdot l_{Pixel} + \epsilon \cdot l_{DWT} \quad (1)$$

The various loss functions used in Super-Resolution Generative Adversarial Networks (SRGANs) include Adversarial Loss, Content Loss, Ranking Loss, PDL Loss, and Pixel Loss. The Adversarial Loss helps the generator produce images that are indistinguishable from real images, while Content Loss ensures that the perceptual characteristics of the generated image are similar to those of the high-resolution image. Ranking Loss measures the divergence between the scores obtained by the Rankers on the classification of the super-resolved image and the high-resolution image. PDL Loss uses the Discrete Wavelet Transform to model complex patterns in images, making the model more robust to spatial variations at different resolutions. Finally, Pixel Loss measures the difference in mean squared error between the pixel values of the super-resolved image and the high-resolution image. This encourages pixel-level similarity between generated and real images. TV Loss is the Total Variation [12] that measures the total amount of variation in an image. More specifically, it evaluates the sum of absolute differences between the intensities of adjacent pixels in an image. This approach favors the generation of images with smoother transitions and fewer abrupt variations between neighboring pixels. The "TV Loss" is the sum of absolute differences between horizontally and vertically adjacent pixels. The use of coefficients α , β , γ , λ , θ , and ϵ in Equation 1 allows to weight the impact of each term,

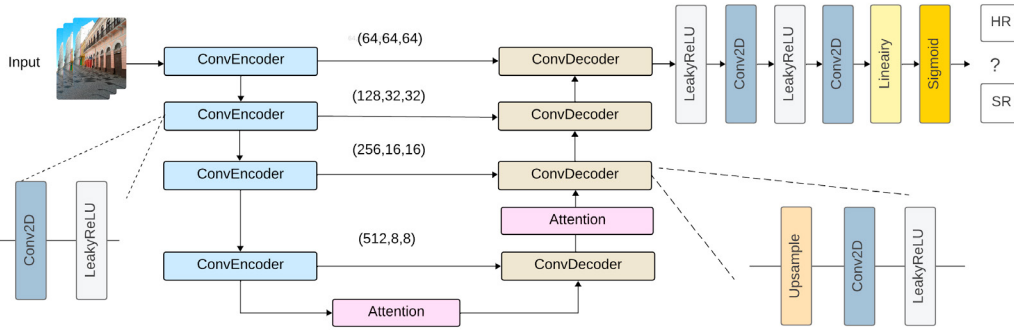


Fig. 3. Discriminator of the SIR-SRGAN-ResNeXt model.

Table 1. The Training and Test Data sets.

Data Set	Description	Original Size (Pixel)	Reduced Size (Pixel)	Sample size	Use
DIV2K [1]	High-resolution image data set	2040 × 1400	128 × 128	900	Training
Flickr2K [17]	High-resolution image data set	2040 × 1356	192 × 192	2650	Training
BSD100 [2]	General image data set	480 × 320	—	100	Testing
Messidor-2 [3]	Diabetic retinopathy image data set	512 × 512	—	250	Testing
URBAN100 [5]	General image data set	984 × 796	245 × 198	100	Testing
Breakhis-400x [16]	Medical image data set	700 × 460	—	176	Testing

thus, providing flexibility to adjust the relative contribution of each regularization in the learning process. Bayesian optimization [15] is used to identify the optimal parameters for our loss function. Bayesian optimization helps to estimate the intervals of promising parameters, then the grid search method is used to select the best parameters. The parameters' performance was measured using PSNR and SSIM after 100 epochs during a convergent and stable training process. Ultimately, the final results are obtained with the following optimal parameters values: $\alpha = 2 \times 10^{-8}$, $\beta = 8 \times 10^{-3}$, $\gamma = 15 \times 10^{-4}$, $\lambda = 3 \times 10^{-2}$, $\theta = 1$, and $\epsilon = 5 \times 10^{-3}$.

4. Experiments

In this section, we present the used data sets, and the validation strategy to train and test the proposed SIR-SRGAN-ResNeXt model, then discuss its performance. The model is evaluated in terms of high-resolution performance using the PSNR, SSIM, LPIPS, HaarPS, and ClipIQA metrics, and also in terms of the output file size and computational cost. Experiments are conducted on four medical and general benchmark image data sets, and a comparison to the state-of-the-art SRGAN models namely SRGAN, RankSRGAN, BSRGAN, SRGAN-ResNeXt, and SIR-SRGAN is performed. Finally, an ablation study is provided to explore the impact the key parameters and components of SIR-SRGAN-ResNeXt have on its performance. All experiments were conducted on a personal computer with an NVidia A100SXM4 GPU and 32GB of memory. The model is trained on 1000 epochs with a batch size of 12 images.

4.1. Data Sets Preparation, Validation Strategy, and Evaluation Metrics

To train and evaluate the proposed model, six data sets were considered: two for training (DIV2K¹ and Flickr2K²) and the remaining four for testing (BSD100³, Messidor-2⁴, URBAN100⁵ and Breakhis-400x⁶). Details and charac-

¹ <https://www.kaggle.com/datasets/joe1995/div2k-dataset>

² <https://www.kaggle.com/datasets/daehoyang/flickr2k>

³ <https://www.kaggle.com/datasets/asilva1691/bsd100>

⁴ <https://www.adcis.net/en/third-party/messidor2/>

⁵ <https://www.kaggle.com/datasets/harshraone/urban100>

⁶ <https://www.kaggle.com/datasets/forderation/breakhis-400x>

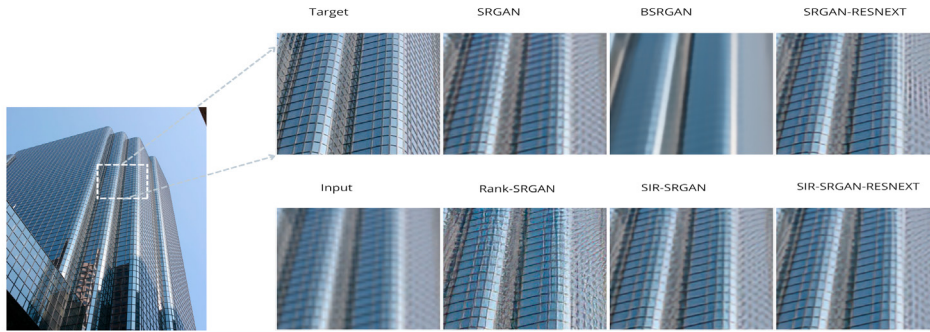


Fig. 4. Visual comparison of the different models on SR ($\times 4$) images from the Urban100 data set.

teristics of each data set are presented in Table 1. To expand the training data set, each image is partitioned into fragments of 192×192 (the crop size) during augmentation. Consequently, for each epoch, a batch-sized sample is randomly extracted from the training set and employed to train the model. The images from the URBAN100 data set are too large to be tested on our computer, hence, they are resized as shown in Table 1. The DIV2K and Flickr2K datasets are used exclusively for training the model. These two data sets provide a total of 3,550 training images. To enhance the robustness of our model and ovoid overfitting, a K-Fold validation strategy is employed, with 10 folds. Hence, each fold yields 3195 images for training and 355 images for validation. After applying the data augmentation mechanism, where each training image is divided by a crop of size 192×192 , a total of 223,650 training images is generated. This approach allows us to maximize the utilization of our resources and optimize the learning potential of our super-resolution model. The proposed model is designed to enhance image resolution, hence, its performance is evaluated on the most relevant metrics to measure image quality and resolution, namely ClipIQA [19], LPIPS [23], HaarPSI [11], PSNR [7], and SSIM [20]. Image storage space is also an essential aspect, thus, the output file size of enhanced images is measured. Finally, the computational time required for each model to process each of the four test image datasets is also evaluated.

4.2. Results

Global visual results on the four test sets are shown in Figures 4 and 5. Visually speaking, all the models return pretty similar SR images except for BSRGAN which generates significantly smoother images referring to less pronounced noise and edginess within the image. The SR ($\times 4$) close-ups on random instances from the four test sets show that SIR-SRGAN-ResNeXt seems to generate high-resolution images with a good definition of edges and texture. A deeper insight into the results is given in Table 2 showcasing the performance of the different models on the four test sets and based on the evaluation metrics presented in Section 4.1. The best measurements are highlighted in bold and the second-best in bold-italic. A first look at the results indicates that, on most metrics, SIR-SRGAN-ResNeXt outperforms competitor SRGANs, namely, in terms of high resolution with an optimal output file size.

ClipIQA is a metric that tells whether a photo is Good or Bad based on visual human perception without a reference image. ClipIQA is a neural network trained on a dataset of images labeled as Good or Bad photos. To have a high clipIQA it is enough for the output image to be as smooth as possible even if it does not come any close to the real reference image. ClipIQA is a metric that takes the SR output image as a single parameter, hence, it does not compare it to any reference image, it just refers to human perception. The other pixel-based metrics namely PSNR, SSIM, LPIPS, and haarPSI take as parameters both the output SR image and the original high-resolution (HR) image and perform a pixel-wise comparison between them. In this regard, both RankSRGAN and BSRGAN give top ClipIQA values, however, the other pixel-based metrics that compare the SR pixels to the original image are better and overall the highest for SIR-SRGAN-ResNeXt.

The URBAN100 buildings data set, as shown in Figure 4, is a particularly challenging one because it encloses multiple high frequencies, mainly shapes with visible borders. The SIR-SRGAN-ResNeXt model performs the best on most pixel-based metrics with a relatively low output file size. In Figure 5, the Breakhis-400x dataset is a data set of cancer scans where images encompass many shapes with visible borders (e.g., cells). Moreover, Messidor-2, a dataset

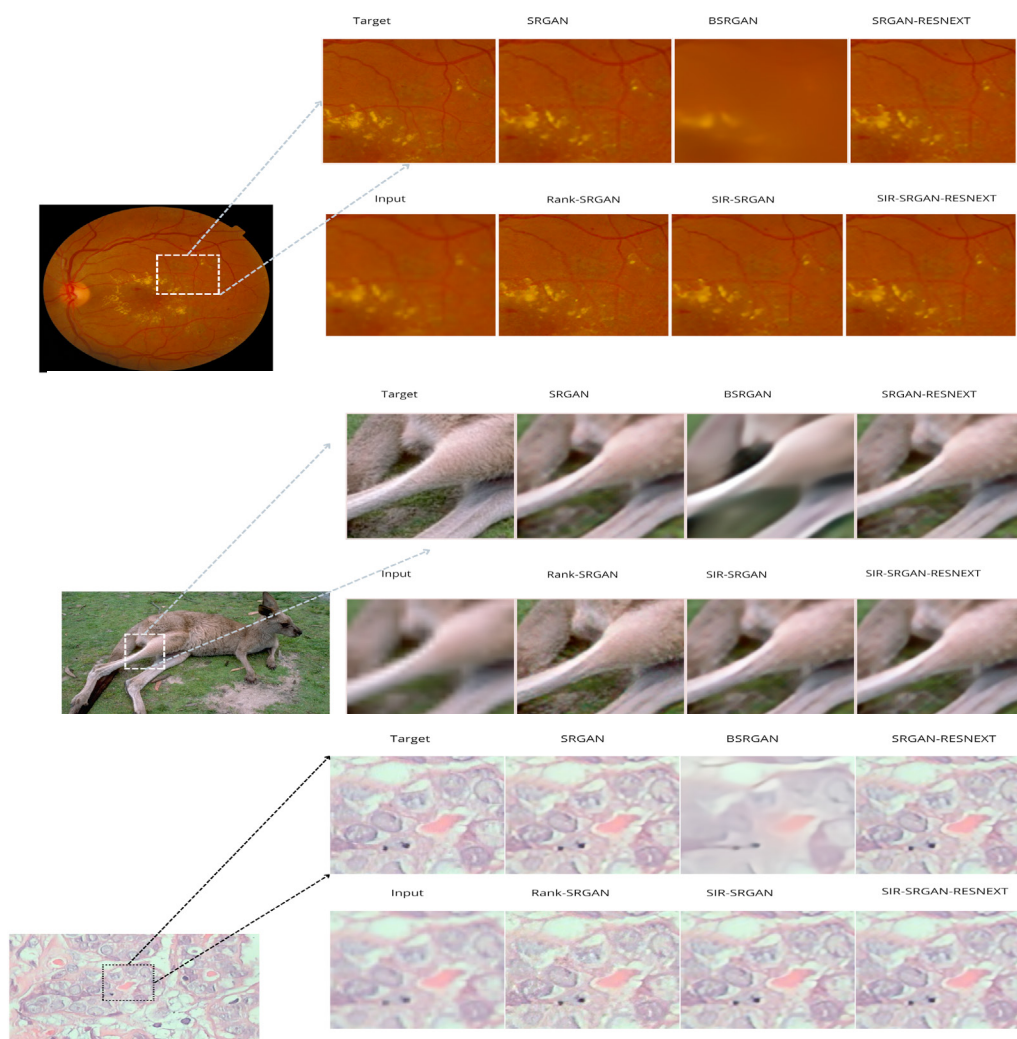


Fig. 5. Visual comparison of the different models on SR (x4) images from the Messidor, BSD1100 and Breakhis-400x data sets.

that consists of fundus images of the eye used for earlier prediction of Diabetic Retinopathy, shows the existence of highly delicate blood veins, crucial to highlight but not visible to the naked eye. On both medical data sets, our model gives the best metrics except for ClipIQA. The image is not considered aesthetic enough from a human visual point of view, but it is the closest to the real image according to the pixel-based metrics.

For the most part, the proposed SIR-SRGAN-ResNeXt model stands out significantly in the super-resolution field, offering favorable performance with a relatively smaller output file size than its competitors. This makes it a particularly good choice for applications handling large quantities of data, where conservation of storage space is crucial. Results also ensure the SIR-SRGAN-ResNeXt outperformance in terms of the pixel-based metrics PSNR, SSIM, LPIPS, and HaarPSI. These measurements testify to the quality of the images generated by the model, underlining its ability to improve resolution while maintaining a modest file size.

Comparatively, the BSRGAN model produces excessively smooth images, impacting its performance in PSNR, SSIM, LPIPS, and HaarPSI. Authors of the BSRGAN paper focused their application on human faces and moving images, which limits the effectiveness of the model on medical images, particularly those related to cancer research and Diabetic Retinopathy prediction. As for Rank-SRGAN, although it achieves a high score in ClipIQA due to the reduction of blur in the images, it was observed that this improvement was accompanied by the removal of crucial small details in the image. In particular, the loss of these details can be detrimental in medical applications requiring careful analysis.

Table 2. Image Enhancement performance comparison on the four test data sets. The best and second-best performances are highlighted in **bold** and **bold-italic**, respectively.

Model	PSNR	SSIM	LPIPS	HaarPSI	ClipQA	Output File Size (MB)	Time (s)
BSD100 data set							
SRGAN	27.517	0.9087	0.0677	0.8874	0.3605	21.0	14
BSRGAN	24.932	0.8296	0.2358	0.8514	0.5699	16.2	15
SRGAN-ResNeXt	27.686	0.9077	0.0675	0.8900	0.3502	21.2	12
RANK-SRGAN	25.054	0.8582	0.0788	0.8484	0.7248	29.7	9
SIR-SRGAN	27.705	0.9084	0.0675	0.8907	0.3683	21.1	13
SIR-SRGAN-ResNeXt	27.816	0.9099	0.0660	0.8929	0.4533	21.0	12
URBAN100 data set							
SRGAN	24.156	0.8690	0.1071	0.8637	0.4066	7.77	5
BSRGAN	21.216	0.7378	0.2746	0.7937	0.4883	6.77	12
SRGAN-ResNeXt	24.353	0.8715	0.0951	0.8696	0.4678	7.90	6
RANK-SRGAN	22.804	0.8420	0.0932	0.8315	0.6002	10.3	6
SIR-SRGAN	24.327	0.8704	0.0997	0.8687	0.4506	7.81	5
SIR-SRGAN-ResNeXt	24.500	0.8746	0.0980	0.8730	0.5132	7.79	6
Messidor-2 data set							
SRGAN	40.730	0.9788	0.0131	0.9888	0.4142	58.8	56
BSRGAN	35.001	0.9266	0.1169	0.9682	0.6588	34.2	46
SRGAN-ResNeXt	40.764	0.9806	0.0144	0.9886	0.3562	60.0	53
RANKSRGAN	39.457	0.9700	0.0167	0.9851	0.4941	82.6	17
SIR-SRGAN	40.882	0.9802	0.0160	0.9896	0.5310	59.4	54
SIR-SRGAN-ResNeXt	42.079	0.9827	0.0121	0.9918	0.4358	60.2	56
Breakhis-400x data set							
SRGAN	35.6487	0.9879	0.0122	0.9628	0.0980	71.5	50
BSRGAN	28.7088	0.9492	0.0970	0.9434	0.4320	66.5	43
SRGAN-ResNeXt	36.4377	0.9884	0.0114	0.9769	0.0860	71.1	49
RANK-SRGAN	33.6930	0.9788	0.0206	0.9499	0.2090	93.1	16
SIR-SRGAN	36.8844	0.9886	0.0110	0.9793	0.1010	70.4	49
SIR-SRGAN-ResNeXt	37.2361	0.9891	0.0101	0.9822	0.1169	70.4	48

The SIR-SRGAN-ResNeXt also stands out for its optimal compromise between reasonable file size and very good performance regarding the various relevant metrics. This performance proved particularly remarkable in tests on the Breakhis-400x images, and image slices related to cancer research, reinforcing its relevance in demanding medical contexts. As a result, we suggest SIR-SRGAN-ResNeXt as a promising solution for super-resolution applications, combining storage efficiency, not the best but relatively reasonable computation time, and high quality for an image, particularly for crucial and meticulous medical images such as cancer slice and fundus of the eye.

4.3. Ablation Study

In the led ablation study, key parameters and components are explored on a maximum of 150 epochs to observe their impact on the performance of the proposed SIR-SRGAN-ResNeXt model. The study evaluates the quality metrics of the generated image and the stability of the model during training.

The first experiment is conducted to find the best discriminator to stabilize the GAN's learning. The experiment is performed using three different discriminators, U-Net, PatchGAN, and ViT, and evaluated on the Urban100 dataset. Results in Figure 6 show that the PSNR and SSIM metrics are more stable when the U-Net discriminator is used compared to the other two discriminators.

Then, the Urban100 dataset images are used to assess the impact the number of attention layers has on the image quality. The study tests up to 3 attention layers and finds, as shown in Figure 7, that PSNR and HaarPSI increase with the number of attention layers, while SSIM and LPIPS decrease beyond 3 layers. The study concludes that the best results are achieved with 2 attention layers.

The ResNeXt model uses cardinality to improve performance by varying the size of the set of transformations. Figure 8 shows that a cardinality of 64 yields the best PSNR and SSIM values, while a cardinality of 16 also yields good results. However, a cardinality that is too low or too high does not give good results. Although a cardinality of

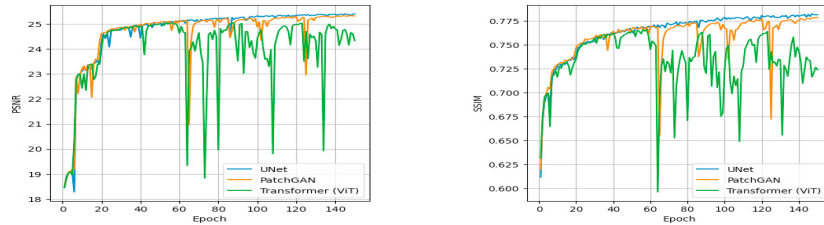


Fig. 6. Evolution of the Stability of the PSNR and SSIM during training with different discriminators.

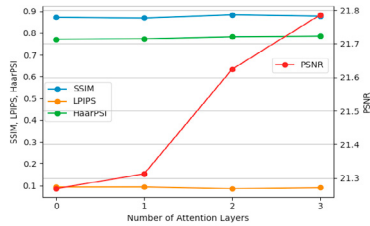


Fig. 7. Metrics evolution regarding the number of attention layers.

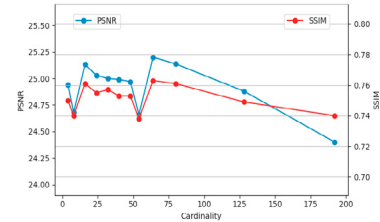


Fig. 8. Metrics evolution regarding the ResNeXt's cardinality.

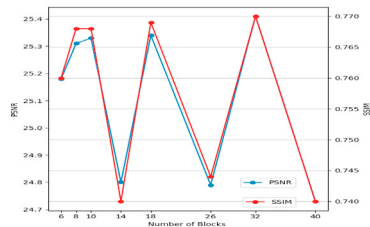


Fig. 9. Metrics evolution regarding ResNeXt Blocks.

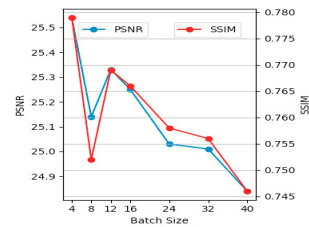


Fig. 10. Metrics evolution regarding the training Batch size.

64 is ideal, it requires a GPU with at least 32 GB of memory for training. Therefore, the model was trained with a cardinality of 16 due to the limited GPU access.

Moreover, the influence of varying the number of ResNeXtBlocks on the quality of generated images is also studied. Different versions of the model are trained over 100 epochs, and it is observed in Figure 9 that PSNR and SSIM are better when the number of blocks is either 8, 16, or 32. A divisor of 64 is required for the number of blocks as the input and output channels of ResNeXt Blocks are 64. The best metrics are achieved with some blocks equal to 32, but the model with 32 ResNeXtBlocks requires a GPU with at least 32 GB of memory for training, while the model with 8 ResNeXtBlocks requires 12 GB of memory.

Finally, the model's training along 80 epochs while modifying the batch size is examined. The Div2k validation dataset is used to perform the tests during training. Figure 10 shows good PSNR and SSIM values with smaller batch sizes (i.e. 4). However, a very small batch size leads to longer training time. Therefore, a batch size of 12 was chosen to converge more quickly towards an optimal model.

5. Conclusion

This paper introduces SIR-SRGAN-ResNeXt, an extended iteration of SIR-SRGAN, as an improved architecture for image super-resolution using two rankers, a ResNeXt-based generator, and a U-Net discriminator. In this model, a ResNeXt-based generator replaces the standard one, to enable learning more complex features and generating higher-quality super-resolved images. The U-Net discriminator, with spectral normalization and attention mechanisms, enhances the model's ability to evaluate the generated images, contributing to better discrimination between the reconstructed and the original images. The loss function integrates various elements with their carefully adjusted coefficients, focusing on perceptual similarity and pixel-level accuracy. The noticeable advantages of SIR-SRGAN-ResNeXt over other super-resolution GAN models are its enhanced learning capability thanks to its ResNeXt generator and U-Net discriminator, and its improved image evaluation with a multi-component loss func-

tion that considers the various image quality criteria. However, the model is limited by execution time, which might hinder its use in real-time applications, and by its hyperparameters requiring careful adjustment for optimal results. To improve the SIR-SRGAN-ResNeXt model, future work could focus on optimizing its computational complexity to reduce its run time and exploring new neural network architectures for the generator such as using personalized attention mechanisms. We also consider adapting the proposed model for other computer vision applications such as image compression.

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